

```
import numpy as np
import pandas as pd
df=pd.read_csv('/content/Mall_Customers.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   CustomerID            200 non-null   int64
 1   Genre                 200 non-null   object
 2   Age                  200 non-null   int64
 3   Annual Income (k$)    200 non-null   int64
 4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
df.head()
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
df.describe()
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
df.Age.mean()
```

```
np.float64(38.85)
```

```
df.Age.min()
```

```
18
```

```
df['Age'].max()
```

```
70
```

```
df.Age.median()
```

```
36.0
```

```
df.Age.std()
```

```
13.969007331558883
```

```
df.Genre.mode()
```

Genre

0 Female

dtype: object

print(df.groupby('Genre').mean())

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
Genre				
Female	97.562500	38.098214	59.250000	51.526786
Male	104.238636	39.806818	62.227273	48.511364

df.shape

(200, 5)

df.dtypes

	0
CustomerID	int64
Genre	object
Age	int64
Annual Income (k\$)	int64
Spending Score (1-100)	int64

dtype: object

summary=df.groupby('Genre')['Annual Income (k\$)'].describe()
print(summary)

	count	mean	std	min	25%	50%	75%	max
Genre								
Female	112.0	59.250000	26.011952	16.0	39.75	60.0	77.25	126.0
Male	88.0	62.227273	26.638373	15.0	45.50	62.5	78.00	137.0

summary=df.groupby('Age')['Annual Income (k\$)'].describe()
print(summary)

	count	mean	std	min	25%	50%	75%	max
Age								
18	4.0	51.250000	14.056434	33.0	44.25	53.5	60.50	65.0
19	8.0	57.000000	20.632845	15.0	47.50	63.5	67.25	81.0
20	5.0	41.600000	24.815318	16.0	21.00	37.0	61.00	73.0
21	5.0	38.800000	19.018412	15.0	30.00	33.0	54.00	62.0
22	3.0	31.333333	22.278540	17.0	18.50	20.0	38.50	57.0
23	6.0	41.500000	23.441416	16.0	20.75	41.5	60.00	70.0
24	4.0	39.250000	16.357975	20.0	33.50	38.5	44.25	60.0
25	3.0	57.666667	29.263174	24.0	48.00	72.0	74.50	77.0
26	2.0	58.000000	5.656854	54.0	56.00	58.0	60.00	62.0
27	6.0	63.166667	18.400181	40.0	49.50	63.5	75.25	88.0
28	4.0	85.250000	11.615363	76.0	76.75	82.0	90.50	101.0
29	5.0	63.600000	28.866936	28.0	40.00	73.0	79.00	98.0
30	7.0	76.142857	39.612408	19.0	56.00	78.0	93.50	137.0
31	8.0	48.375000	23.323425	17.0	35.50	41.5	70.50	81.0
32	11.0	87.181818	26.809768	48.0	74.00	77.0	100.00	137.0
33	3.0	80.333333	35.837597	42.0	64.00	86.0	99.50	113.0
34	5.0	79.000000	15.968719	58.0	78.00	78.0	78.00	103.0
35	9.0	46.666667	38.632888	18.0	21.00	24.0	74.00	120.0
36	6.0	81.000000	22.556595	37.0	85.50	87.0	87.00	103.0
37	3.0	65.000000	40.112342	20.0	49.00	78.0	87.50	97.0
38	6.0	74.500000	20.462160	54.0	64.75	69.0	76.25	113.0
39	3.0	72.666667	4.725816	69.0	70.00	71.0	74.50	78.0
40	6.0	61.666667	19.551641	29.0	55.50	64.5	70.50	87.0
41	2.0	101.000000	2.828427	99.0	100.00	101.0	102.00	103.0
42	2.0	60.000000	36.769553	34.0	47.00	60.0	73.00	86.0
43	3.0	65.666667	15.695010	48.0	59.50	71.0	74.50	78.0
44	2.0	75.500000	3.535534	73.0	74.25	75.5	76.75	78.0
45	3.0	69.333333	50.767444	28.0	41.00	54.0	90.00	126.0
46	3.0	59.000000	36.755952	25.0	39.50	54.0	76.00	98.0
47	6.0	70.166667	27.694163	43.0	51.75	65.5	76.25	120.0
48	5.0	58.200000	13.700365	39.0	54.00	60.0	61.00	77.0
49	7.0	51.000000	12.884099	33.0	40.50	54.0	62.00	65.0
50	5.0	58.600000	18.420098	40.0	43.00	58.0	67.00	85.0

51	2.0	55.500000	16.263456	44.0	49.75	55.5	61.25	67.0
52	2.0	55.500000	45.961941	23.0	39.25	55.5	71.75	88.0
53	2.0	39.500000	9.192388	33.0	36.25	39.5	42.75	46.0
54	4.0	59.750000	30.998656	28.0	42.25	55.0	72.50	101.0
55	1.0	57.000000	NaN	57.0	57.00	57.0	57.00	57.0
56	1.0	79.000000	NaN	79.0	79.00	79.0	79.00	79.0
57	2.0	64.500000	14.849242	54.0	59.25	64.5	69.75	75.0
58	2.0	54.000000	48.083261	20.0	37.00	54.0	71.00	88.0
59	4.0	65.250000	21.792583	43.0	51.25	62.5	76.50	93.0
60	3.0	43.333333	11.547005	30.0	40.00	50.0	50.00	50.0
63	2.0	56.500000	12.020815	48.0	52.25	56.5	60.75	65.0
64	1.0	19.000000	NaN	19.0	19.00	19.0	19.00	19.0
65	2.0	50.500000	17.677670	38.0	44.25	50.5	56.75	63.0
66	2.0	63.000000	0.000000	63.0	63.00	63.0	63.00	63.0
67	4.0	45.500000	18.699376	19.0	40.00	50.5	56.00	62.0
68	3.0	56.666667	7.767453	48.0	53.50	59.0	61.00	63.0
69	1.0	44.000000	NaN	44.0	44.00	44.0	44.00	44.0
70	2.0	47.500000	2.121320	46.0	46.75	47.5	48.25	49.0